# Verification Results and v1.0 Comparison

**Verification of recorded permission and protected dataset packages**

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## Abstract

Preparing a dataset does not, by itself, give permission to use it. APOD-TR-010-R examines how Q-Pathformer Enterprise v2 checks recorded permission and protects prepared data before an operation proceeds. It presents the verification results for the method described in APOD-TR-010 and compares them with the earlier v1.0 lifecycle.

The current study uses a small synthetic dataset. A permitted package completes a dry run without training a model. Deliberately invalid cases, including the wrong operation and altered package contents, are rejected.

These findings support a bounded conclusion: the tested controls distinguish a prepared dataset from a package permitted for a named operation. They do not establish better model quality, production readiness or legal entitlement. Earlier training and governance results are included for comparison, not presented as new outcomes of this study.

TR010 serves a dual role: an implementation report explaining the method and its governance boundaries, and a reusable reference protocol for evidence-bound state transitions. TR010-R shows what the verification establishes and where its limits remain. Legal review and organizational authorization remain separate human responsibilities. The study does not demonstrate automated legal approval.

**Keywords**

Governed datasets; recorded permission; fail-closed controls; package integrity; verification; provenance; standards exploration.

**History and provenance**

The companion History and Provenance Notice (history_notice.md) records the chronology and evidence boundaries. Detailed source identities are provided in Appendix A and the preserved ledgers; the preceding companion identity is recorded in `TR010_R02_EDITORIAL_HANDOFF.json`. Historical records retain their original scope. [E01, E02]

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# 1 Purpose and scope

## 1.1 Purpose and relationship to TR010

APOD-TR-010 explains the governed dataset method: prepared data, recorded permission and package integrity are checked separately before a named operation can proceed. APOD-TR-010-R examines the evidence for those checks. Its purpose is to show which behavior was observed, under which conditions, and what cannot be concluded from it. [E01]

The comparison does not characterize v1.0 as a failed release. V1.0 established a three-path lifecycle and reviewable execution examples. The v2 work makes authorization and integrity boundaries more explicit and tests cases in which technical eligibility or historical metadata could otherwise be interpreted too broadly. A change in control semantics is not, by itself, evidence of a better trained model. [E02, E10]

## 1.2 What the evidence shows

The evidence has three time periods. September checks examine the current controls using synthetic data. July records concern preparation of an earlier v2 candidate. The v1.0 records show earlier lifecycle and model-run examples. Keeping these groups separate prevents an old success from being presented as a new result. Documentation records identify the files; they are not additional experiments.

**Table 1 Evidence class and permitted interpretation**

| Evidence class | Recorded period | Permitted interpretation |
|---|---|---|
| Historical v1.0 execution | April 2026, summarized in July | Earlier lifecycle, onboarding, model-run and qualified CSC observations. |
| Historical v2 release verification | July 2026 | Tests, export and integrity checks for the identified historical candidate. |
| Current local control verification | 5 September 2026 | Recorded tests and synthetic rehearsal for the separately captured development source. |
| Documentary preparation | Initial preparation on 5 September; requirements, operator-report and edition supplements on 6 September 2026 | Documentary reconciliation and source identity checks, not additional runtime evidence. |

## Edition scope

Q-Pathformer Enterprise v2.0 Standards Edition is the current compact standards-review edition. APOD-TR-010 explains the implemented control method and reusable evidence-bound transition protocol. APOD-TR-010-R reports the bounded September synthetic verification and separately identified historical comparisons. The v2 edition does not reproduce the complete v1.0 enterprise lifecycle. The planned v3 enterprise release is intended to combine the v1.0 three-path foundation with the v2 authorization and package-integrity controls for end-to-end reproduction within a defined enterprise scope. V3 is not

released, and its intended outcomes are not verified v2 results. Future CSC Assurance Record work remains separately scoped; it is not made operational by publishing these papers.

### 1.3 Source availability and strength

Appendix A and its source ledger identify the records used for the findings. File paths and hashes allow a specific record to be located and compared. They identify the evidence examined; they do not independently prove every statement within it.

The September unit log and subprocess rehearsal are retained locally. The July M06, corrected M10, corrected M11, M12 and CSC qualification records are also retained. Some original v1.0 run artifacts and datasets are not included in this compact workspace. Accordingly, Chapter 4 describes values reported by the retained July comparison source, not a fresh reopening or execution of every original v1.0 artifact. The historical evidence directory itself is an explicit placeholder rather than a populated replay dataset. [E03, E04, E10-E15, E22]

References such as [E03] identify the local records in Appendix A. Ranges denote the listed source groups, not an assertion that every group supports every sentence. Implementation inspection supplies context; executed test logs supply the reported test outcomes.

### 1.4 Scope of the study

The current verification is a synthetic control test, not a new model-training or full three-path experiment. No new adapter or CSC score is reported. Historical records retain their original identities, and no legal engagement is established by the worked example. Document checks are separate from runtime tests.

The scope is standards exploration: a method and an evidence set that can be examined and reproduced within their stated limits. Filenames containing release, certificate or frozen do not establish public availability, certification or current authority. Publication and real-world use require their own decisions. [E02, E19]

### 1.5 Demonstration and background

**Synthetic dataset -> one-operation authorization -> sealed package -> valid dry run and rejection tests -> verification report.**

Four made-up records and a prepared STATE_1/freeze/registry fixture provide the input. One-operation means one named authorized scope, not one-time use. Separate signing roles protect the recorded decision and package inventory. The permitted package completes a dry run; selected wrong-operation, routing and tampering cases are expected to reject. The report preserves these outcomes rather than presenting a new model. [E04, E07]

TR007 supplies lifecycle background, TR008 explains signal exchange, and TR009 records session-bound governance observations. TR010 explains the current method and reusable protocol. This research progression is not an execution sequence. The executable flow is separate from the historical July milestone map in Appendix B; none of their arrows supplies permission. [E01, E26]

# 2 Current September verification

## 2.1 Environment and source binding

The recorded environment identifies Python 3.11.9 on `Windows-10-10.0.26200-SP0`. Its eight required package pins matched the installed versions recorded in that environment. The environment record belongs to the September local preparation; it is not a claim about every machine on which the repository may later run. [E05]

The September source was based on private-repository parent HEAD `57c7fbbed476a8f74ba4ca7b6acb495791ba5524` with uncommitted changes. That commit is a parent reference, not a complete identity for the tested working tree. The source inventories and rehearsal hashes provide the additional byte-level binding. [E06]

**Table 2 Recorded dependency and pinned version**

| Recorded dependency | Pinned version |
|---|---|
| attrs | 26.1.0 |
| cffi | 2.1.0 |
| cryptography | 49.0.0 |
| jsonschema | 4.26.0 |
| jsonschema-specifications | 2025.9.1 |
| pycparser | 3.0 |
| referencing | 0.37.0 |
| rpds-py | 0.30.0 |

The retained consolidation rehearsal was recorded at `2026-09-05T04:41:42.010105+00:00`. Its source map contains 11 implementation, harness and backend-profile files. A separate read-only source comparison on 6 September 2026 found all 11 local hashes equal to their recorded hashes; that comparison was retained in the compact-revision handoff. This narrows the connection between that rehearsal and the inspected files, but is not a complete dependency attestation or a trusted timestamp. The broader source capture remains necessary to identify the working selection. [E04-E06]

## 2.2 Automated test outcome

The retained consolidation log records 48 tests in 2.585 seconds: 47 passed, one skipped and zero failed. The elapsed time is a test-log observation, not an inference or training-performance benchmark. [E03]

**Table 3 Test module area and recorded outcome**

| Test module area | Tests | Recorded outcome |
|---|---|---|
| Deterministic release export | 8 | 7 passed; 1 skipped |
| Fail-closed lifecycle | 7 | 7 passed |
| Complete package sealing | 5 | 5 passed |
| Private document allowlist | 7 | 7 passed |
| Single-scope contract | 15 | 15 passed |
| Training authorization and migration | 6 | 6 passed |
| Total | 48 | 47 passed; 1 skipped; 0 failed |

These counts were derived from the named entries in the retained log, not from counting source functions as executed tests. Several tests exercise multiple cases, but those cases are not inflated into additional test counts. A passing test count is not a percentage of all possible behaviors or a measure of production coverage.

The skipped case was the exporter's symbolic-link rejection test, whose environment guard reported that symbolic-link creation was unavailable. It remains a skip in that retained run. The distinct Windows short-name temporary-path regression and the path containing .. both appear as executed passes in the same log. These tests address path spelling and fixture behavior; they do not erase the separate symbolic-link skip. [E03, E21]

The separate targeted observations and Windows capability requirements are consolidated in Appendix C.4. Neither the later skip nor the operator-reported single-test pass changes this retained full-suite result. [E25, E27]

## 2.3 Controls supported by the tests

The authorization contract is `qpath_training_authorization_v2.2`. An ALLOW artifact has exactly one explicit scope; a DENY artifact has none. Preparation, copy migration and the local research runner are bound to `validation_parameter_update`. A caller-supplied scope label cannot turn those endpoints into production training, external-provider training, model release or deployment. [E07]

**Table 4 Control area and bounded conclusion**

| Control area | Executed evidence | Bounded conclusion |
|---|---|---|
| Scope cardinality and version | Builder, schema and verifier rejection cases | Missing, multiple, duplicate or unknown ALLOW scopes, and unsupported schema versions, are rejected. DENY contains no authorized scopes. |
| Endpoint binding | Preparation, migration and runner mismatch cases | Relabelling the request does not select another operation. |

| Control area | Executed evidence | Bounded conclusion |
|---|---|---|
| Decision and validity | Signed DENY and time-boundary cases | A valid signature does not turn denial or an invalid validity interval into permission. |
| Dataset and registry binding | Signed mismatch cases | Tested version, digest and registry mismatches are rejected. |
| Role separation and tampering | Reused-key, signature and inventory cases | Tested key reuse and changed package content fail the required checks. |
| Rejection outputs | Route and rejected-run assertions | Tested rejections do not enter the backend or leave a final success directory. |

The named validity-boundary test checks an inclusive start and an exclusive expiry. Source inspection also confirms a 31-day maximum active interval, measured from not-before to expiry; no dedicated over-31-day assertion was identified in the two inspected authorization test modules. The latter is an implementation observation, not an added executed test. The runner tests also cover a correctly sealed package with the wrong operation scope. Those cases matter because a well-formed, authentically signed object can still be ineligible for a particular endpoint. This report does not equate successful cryptographic verification with operational permission. Preparation reads a supplied local registry and hashes the frozen source; the runner checks recorded sealed-manifest bindings rather than querying a live registry or rehashing the original source at use time. TR010 sections 3 and 4 explain these implemented boundaries. [E03, E07]

### 2.4 Nine-step synthetic CLI rehearsal

The separate subprocess rehearsal used synthetic STATE_1 and registry fixtures, disposable keys and a four-record dataset. It did not obtain its initial state by executing all three lifecycle paths. The following sequence is recorded in the retained report. Exit code 1 is the expected outcome of four negative cases, not a failed overall rehearsal. [E04]

**Table 5 Step and expected and observed exit**

| Step | Operation | Expected and observed exit |
|---|---|---|
| 1 | Generate authorization-signing key pair | 0 |
| 2 | Generate separate package-sealing key pair | 0 |
| 3 | Create one-scope authorization | 0 |
| 4 | Reject multiple ALLOW scopes | 1 |
| 5 | Prepare authorized and sealed training package | 0 |
| 6 | Execute valid local dry run | 0 |
| 7 | Reject the wrong endpoint operation | 1 |
| 8 | Reject unsupported package/log routing | 1 |
| 9 | Reject package tampering | 1 |

All nine expected outcomes matched. The recorded negative error identifiers were `allow_requires_exactly_one_scope`, `endpoint_operation_scope_mismatch`, `unsupported_training_package_path_for_top_level_run_log`, and `package_inventory_verification_failed`. The long identifiers are diagnostic labels, not separate lifecycle states.

The report records an unchanged source dataset and synthetic registry, absent rejected-run directories, disposable keys, and an enabled guard against model imports and socket connections. Its successful run artifacts include `model/model_stub.json`. That stub is dry-run evidence, not a trained model. The temporary workspace was removed after the rehearsal; the report retains its hashes and outcomes, not the temporary private keys. [E04]

### 2.5 Interpretation and comparison with earlier September checks

The earlier Windows investigation recorded 33 tests with 32 passes and one skip. The later single-scope work expanded the suite to 48 tests. These are sequential snapshots, not 81 independent tests of one final artifact. The reported Windows failure concerned an unresolved temporary fixture root being compared with a resolved runtime path. The bounded fix resolved the fixture root and added regression coverage; it was not a cryptographic failure. [E03, E21]

The historical investigation also identified an empty-directory side effect on unsupported log routing. The later September slice moved route resolution ahead of final run-directory creation and added rejection assertions. The current report cites those later tests without rewriting the earlier finding. No hosted CI rerun is established by the retained local consolidation evidence. [E04, E07, E09, E21]

# 3 Historical July release verification

## 3.1 Distinct historical candidate

The July records refer to source commit `cf9721afb81273d82667dd7b33ee5e5f3ad806d9`, source tag `v2.0-qpathformer-enterprise-source-baseline`, and source tree `866cb4246d4fdc5e31c3a8be32249dc1b00b60ec`. These are not the September parent reference or the identity of this report. The term corrected candidate below always refers to the July M11 object. [E12, E13]

## 3.2 Deterministic export and historical preservation

The retained M06 record and July comparison source report two separately generated export targets with matching file sets and hashes. Historical-byte preservation and controlled active-text normalization were distinct export treatments. They are not a general permission to normalize arbitrary evidence. [E10, E11]

**Table 6 Historical export measure and recorded result**

| Historical export measure | Recorded result |
|---|---|
| Separate M06 export targets | 2 |
| Files per target | 320 |
| Bytes per target | 35,414,938 |
| Cross-target file or hash mismatches | 0 |
| Historical files preserved byte-for-byte | 174 |
| Active text files normalized under policy | 146 |
| Extra or missing files | 0 |
| Source mutation detected | No |

### Historical deterministic target digest

`62a866976506a65b9e238faa5e9e94d6f32c8991f1b2795a4561fdd3e6ed3080`

The 6 September documentary check inspected the retained records, not the two original export directories. The reported determinism is scoped to their allowlist, normalization rules, implementation and environment. Reproducing a byte identity does not establish that the selected content is lawful, correct or complete for every use.

## 3.3 Corrected M10 and M11 results

The corrected M10 record reports 24 tests: 23 passed and one environment-guarded symbolic-link test was skipped. It also records 23 of 23 release-conformance checks, a passing dependency check, five of five finding-specific assertions, no modified historical evidence files and no unexplained warnings. These are July checks, not the current 48-test result. [E12]

**Table 7 Corrected July candidate measure and recorded value**

| Corrected July candidate measure | Recorded value |
|---|---|
| Publication-inventory entries | 348 |
| Listed pre-record bytes | 35,619,880 |
| Final checksum entries | 350 |
| Final regular files | 351 |
| Final bytes | 35,754,807 |
| Post-checksum target writes | 0 |

### Historical publication-inventory SHA-256

`bed0957ddb1de0b3ed1b76c295f7febc12f7ca53880d7c1fd408996c2bcc07e6`

### Historical checksum-file SHA-256

`47a04a06dcc3595402c95f8028c0cf222ccc09351040b9a3fb76f399e87dbf75`

### Historical final candidate tree SHA-256

`9682ef298961b5288d3c7a59134b6273511f1a066a705d221af71ce69d225bb0`

The publication inventory preceded the build report and checksum file. The checksum set excluded itself, which explains why its 350 entries differ from the final 351-file count. The corrected M11 report records two separately recorded read-only verifications and zero post-checksum writes. The retained inventory and checksum-file bytes can be identified locally, but the compact workspace is not the entire frozen tree they describe. This report does not claim to have reverified every historical member. [E13, E23]

## 3.4 Findings and dispositions remain part of the record

The initial M12 review was not an unqualified PASS. It identified a custom SPDX metadata problem, remote pipe-to-shell installer guidance and mutable GitHub Actions references. Bounded remediation corrected those three findings and reran M10 before rebuilding M11. The substantive licence terms were not changed by the SPDX metadata correction. [E12-E14]

The corrected M11 status was `READY_FOR_M12_RECONSIDERATION`; its next human review was not authorized in that record. A technical correction did not itself supply a new human release disposition. TR010-R preserves that boundary instead of declaring all historical stages complete or restarting M00 onwards. [E13, E14]

M09 also retained its earlier HOLD before recording a separately authorized canonical-equivalence resolution. Its nine-entry historical verification reported three byte-exact entries and six entries reconciled through the sole recorded CRLF-to-LF comparison, while preserving physical source bytes. That historical treatment must not be confused with byte equality or applied silently to new materials. The current local standards subset has its own selection record and is not a claim that every historical vendor member is present. [E16]

### 3.5 Later July documentary context

The main paper and `history_notice.md` separately describe a 21 July pre-rebuild rights/provenance review and proposed canonical-copy selection, remapping and regeneration. These are not results of the corrected 18 July M11 object. No later executed rebuild, new M12 disposition or practitioner legal opinion is established by that documentary summary. The original detailed file-level review set was not established in the inspected controlled repository; its summary remains attributed context rather than a newly authenticated finding. [E02, E19; TR010 section 7.5]

# 4 Historical v1.0 execution results

## 4.1 Source and interpretation boundary

This chapter summarizes the retained July v1.0 comparison source. The observations concern historical runs, predominantly the April release working set. The compact standards workspace does not contain every original dataset, model output or run log; the chapter therefore reports the historical source's findings rather than claiming a new three-path reproduction. Where independent reinspection of original artifacts is needed, the recorded source identity must be used to retrieve the correct version. [E10, E22]

## 4.2 Path 1 corpus and local LoRA

The historical Path 1 release record uses `checkpoint_v4_strict` and package `QPDS-TR-STRICT-ID3-001`. The retained source reports the following preparation counts. [E10]

**Table 8 Historical Path 1 measure and reported value**

| Historical Path 1 measure | Reported value |
|---|---|
| Documents | 16 |
| Extracted signals | 846 |
| Source-signal dataset rows | 1,241 |
| Training rows | 1,117 |
| Validation rows | 124 |

Signals and dataset rows are different units. The increase from 846 signals to 1,241 source-signal rows must not be described as additional independently collected observations. Dataset construction and filtering determine the resulting training and validation rows; their counts are not quality scores.

The historical TinyLlama 1.1B local LoRA run is reported as completed on CUDA, with aggregate training loss approximately 3.4113 and final logged loss approximately 3.4135. The aggregate and last-step values answer different questions and should not be interchanged. The source establishes recorded execution continuity, not a held-out evaluation, improved generalization or a v2 training outcome. This report does not calculate a validation loss from the existence of 124 validation rows. [E10]

## 4.3 Path 2 source-first and runtime options

The source-first option reports four grouped scenarios and four candidates progressing through review, promotion, handoff, freeze, registry handling and training-package preparation. Its historical TinyLlama 1.1B run used four records, with aggregate training loss approximately 3.7577 and final logged loss approximately 3.2855. Repeating four records for the reported 100 epochs demonstrates that the training loop completed; it does not establish useful generalization. [E10]

The runtime option is a different input route. The retained source describes local `gpt-oss:20b` signal generation and a package held below the four-candidate promotion threshold.

It is not evidence that the runtime prompt read the source-first `01_source` folder. Candidate totals across different sessions are not a substitute for the count in the package actually reviewed. A below-threshold HOLD is a useful negative example, not a failed source-first result. [E10]

The runtime model and fine-tuning model also have different roles. `gpt-oss:20b` is reported as a signal emitter in that option; TinyLlama is the reported local LoRA target. The September synthetic rehearsal invoked neither model. Their names do not imply that the Q-Pathformer model architecture itself was trained or benchmarked by the current verification suite. [E04, E10]

### 4.4 Path 3 client JSON and freeze-bundle exception

For the client JSON option, the historical source reports one source file normalized to four records with hash status `ok`, remaining in STATE_3. That is a contained onboarding result, not evidence of promotion, training permission or completed model training. Its positive classification in the July candidate does not mean that this later compact workspace contains the original populated client package. [E10, E22]

For the freeze-bundle option, the earlier ingestion record reported six normalized records and hash status `ok`. The retained July comparison source reports a later audit finding that the files at the identified historical tag did not satisfy their own `hashes.txt`; it excludes that bundle from positive evidence. Both observations are preserved: a reported successful ingestion message and a later reported integrity discrepancy. [E10]

The 6 September compact-revision documentary check did not reopen the complete tagged bundle, recalculate its member hashes, diagnose a new cause or repair the source. The discrepancy remains a disclosed historical finding that requires the original file-level evidence for independent confirmation. Neither the positive client JSON example nor canonical equivalence in a different standards package resolves it.

# 5 Historical CSC evidence and qualifications

## 5.1 Origin and selected evidence

M08 records selection of 25 historical artifacts across four profiles from the identified v1.0 source. Its recorded source commit is `0fa94ecf47d698a894bc868545f110570d0dec50` at tag `v1.0-qpathformer-enterprise`. The qualification record binds these CSC artifacts to Path 2 run `RTRUN-llm-runtime-registration-check-20260413T095208Z-77a538bc`, dated 13 April 2026. Path 1 and Path 3 are contextual, not the originating CSC runs. [E15]

The selected artifact count belongs to the July qualification record. It is not a count of newly generated v2 scores or a representation that all 25 original artifacts have been copied into this compact standards workspace. The retained manifest and qualification report identify the historical selection.

## 5.2 Qualified historical scores

All values below are historical, evidence-limited observations, not current assurance ratings. Finance and Healthcare were mapper-driven v1.0 baselines. Public and AI/emerging technology were curated standards examples, not comparable mapper-driven enterprise baselines. Coverage must be read alongside the number, not as a footnote that can be omitted. [E15]

**Table 9 Historical profile and observed catalogue metrics**

| Historical profile | Composite | Historical level | Observed catalogue metrics |
|---|---|---|---|
| Finance | 91.9 | Provisional 5 | 16/60, approximately 26.7% |
| Healthcare | 84.6 | Qualified 4 | 12/60, 20.0% |
| Public | 93.0 | Curated 5 | 12/60, 20.0% |
| AI and emerging technology | 82.9 | Curated 4 | 12/60, 20.0% |

Finance's qualification identifies provisional saturated governance-chain factors f6.1 and f6.5. Healthcare is a narrow result with sparse observation, not a general healthcare assurance claim. The Public record reports sector validation in its historical artifact but lacks a dedicated mapper-driven baseline. The AI/emerging-technology record explicitly states that sector validation was not met. [E15]

The qualification record requires confidence and evidence provenance to accompany interpretation. This report does not derive a common confidence interval, a direct-telemetry percentage or a proxy percentage from the observed-metric count. Those quantities are not interchangeable. The compact summary does not establish a harmonized, calibrated confidence measure for comparison across sectors. A reader requiring factor-level conclusions must inspect the referenced historical manifest and original evidence before relying on them.

### 5.3 No cross-version scoring claim

The separate governance-snapshot contract is marked `DESIGN_BASELINE_NOT_YET_ENFORCED`. It distinguishes version-bound evidence, observation status, historical non-retroactivity and future calibration. It does not establish an active v2 scorer or a newly completed CSC Assurance Record. [E16, E17]

Consequently, mixing older and newer records does not automatically justify either increasing or decreasing a CSC score. Such a result would require an implemented, versioned rule for eligibility, ageing, coverage, confidence and weighting, with its own validation. TR010-R reports no cross-version blended score. The retained v1a schema names and certificate-like artifacts are historical terminology, not a new certification claim by this report.

Historical naming note: the supplied public-reference review records enterprise v1.0 usage as "Controlled Signal Certificate", derived from Access-PoD "Compliance Star Certificate". Original schema names and artifact titles are retained; neither name establishes certification by this report.

# 6 Comparison between v1.0 and v2

## 6.1 Functional changes rather than model benchmarks

V1.0 demonstrated a governed dataset flow. The v2 work separates the objects and checks that determine whether a particular operation may proceed. The main comparison is therefore one of implementation semantics and evidence handling, not a claim that v1.0 and v2 were evaluated on the same prediction task. [E01, E07, E10]

**Table 10 Area and current v2 treatment**

| Area | v1.0 evidence | Current v2 treatment |
|---|---|---|
| Lifecycle | Three-path execution and HOLD examples | Explicit fail-closed checks and bounded negative outputs. |
| Training permission | Lifecycle and historical metadata | Separate signed authorization with one ALLOW scope and no DENY scopes. |
| Endpoint operation | Earlier execution continuity | Research endpoints enforce their own operation rather than trusting a caller label. |
| Package integrity | Earlier manifests and hash records | Complete package inventory and a separately signed seal, reverified at use. |
| Migration | Historical packages retained | Newly authorized sealed copy; prior signed bytes are not edited in place. |
| Evidence interpretation | Earlier runs and CSC values | Date, source identity, coverage and limitations remain attached to each claim. |

Package sealing records regular member paths, sizes and SHA-256 values and authenticates the deterministic inventory binding with a separate Ed25519 key. Exactly the root-relative `package_seal.json` is excluded from the inventory; a nested file of that name is not excluded by the same rule. TR010 section 4 states the exact signed representation and Merkle rules. The tested authorization and sealing key fingerprints must differ. Two keys establish a technical role distinction; they do not prove two independent organizations or reviewers controlled them. [E07]

Legacy v2.1 authorization artifacts are not silently accepted by the current v2.2 contract, even if an older artifact happened to contain one scope. The recorded migration approach requires a fresh decision, signature and sealed copy. This is not permission to edit an existing signed JSON object or to attach an old signature to new content. [E07, E08]

## 6.2 Comparisons that are not supported

The July source reports different sizes for the broad v1.0 source, v2 source and allowlisted candidate. Those selections contain different asset classes. A smaller release folder is not a smaller or more accurate trained model, and Git blob counts are not directly equivalent to

regular-file counts in a selected export. This report therefore does not use file-count reductions as an optimization score. [E10]

The 24-test July suite and 48-test September suite also differ in scope and include document-allowlist and later contract tests. The larger count demonstrates additional named checks, not twice the security assurance. The repeated four-record historical LoRA run is not statistically comparable with the 1,117-record Path 1 run merely because both used TinyLlama. No common held-out metric or controlled quality comparison is reported. [E03, E10, E12]

### 6.3 Claim and evidence map

**Table 11 Claim and required limit**

| Claim | Evidence basis | Required limit |
|---|---|---|
| The tested v2.2 ALLOW contract is single-scope | E03 and E07 | One scope is not single use or replay prevention. |
| The rehearsal rejects wrong operation, route and tampering | E04 | Synthetic cases, not universal side-effect freedom. |
| July exports were reported deterministic | E11 and E13 | Their original source and candidate identities only. |
| v1.0 local model loops completed | E10 | Historical reported execution, not model quality. |
| Historical CSC artifacts were qualified | E15 | Sparse observations and profile-specific limitations remain. |
| A legal-review workflow is illustrated | E18 and E19 | Fictional record; no actual practitioner approval or runtime credential check. |

The reference control implementation's contribution is the enforceable technical boundary between recorded evidence and a permitted endpoint operation. Human authority, rights assessment and independent assurance remain outside what these tests establish. That division is a design constraint, not a weakness to be hidden by a composite score.

# 7 Reproduction and reviewer procedure

## 7.1 Select the prepared local environment

The existing `docs/execution/TR010_STANDARDS_REFERENCE_RUNBOOK.md` remains the executable reference for the complete prepared local workspace. This section clarifies interpreter selection without changing the scripts, dependency pins or outcome contract. The current route uses PowerShell 7 and Python 3.11 with the eight lightweight pins in section 2.1. It does not read the old populated `01_source` examples, call Ollama, download TinyLlama or calculate a CSC score. [E08]

Use the actual executable from an already prepared isolated environment. An explicit `py -3.11` selector is not by itself a binding to an activated environment. The Python launcher documentation distinguishes these behaviors. No particular `.venv` path is assumed below. The commands are reproduction guidance, not operations executed during this manuscript revision. [W1]

```
$Workspace = Read-Host 'Full path to the prepared local workspace'
Set-Location -LiteralPath $Workspace
$Python = (Resolve-Path -LiteralPath (
    Read-Host 'Full path to the prepared environment python.exe'
)).ProviderPath
& $Python -B -c @'
import sys
print(sys.executable)
print(sys.version)
print('Environment prefix:', sys.prefix)
assert sys.version_info[:2] == (3, 11), 'Python 3.11 required'
'@
if ($LASTEXITCODE -ne 0) { throw 'Interpreter check failed.' }
```

The printed executable, version and environment prefix identify the selection; they do not install or authenticate its dependencies. Compare the prepared environment with the recorded pins before running project checks. The source harness invokes child processes through `sys.executable`, so using the same `$Python` for the harness also selects that interpreter for its children. [E05, E07, E08]

## 7.2 Internal preparation is not public pack verification

`tools/verify_private_workspace.py` intentionally checks a private working-draft allowlist. The inspected selection requires 11 identified DOCX files and rejects PDFs, ZIPs and unselected documents. It is not a generic acceptance test for a two-PDF public standards pack. Do not rename it, remove its requirements or weaken its allowlist to obtain a cosmetic pass. [E07-E09]

Only for that corresponding complete private workspace, the internal check is:

```
& $Python -B .\tools\verify_private_workspace.py
if ($LASTEXITCODE -ne 0) { throw 'Private workspace check failed.' }
```

Close Word files in that workspace before this strict internal scan because a temporary `~$` owner file is not an allowlisted manuscript. This warning does not authorize deletion of unrelated files and does not turn a transient document lock into a governance finding.

The documented reproduction route applies to the complete prepared local workspace, not to the two papers alone. The following unit and rehearsal commands require its scripts, schemas, tests, fixtures, backend profile and relevant provenance. A public-pack-only reproduction route is not established by the retained local evidence. Missing files must be reported, not invented or silently bypassed.

### 7.3 Unit and synthetic rehearsal commands

Continue in the same PowerShell session with the selected `$Python` and workspace. Preserve the `-B` intent and exit-code checks. Use a unique report path and do not overwrite retained evidence.

```
& $Python -B -m unittest discover `
  -s .\tests\unit -p 'test_*.py' -v
if ($LASTEXITCODE -ne 0) { throw 'Unit checks failed.' }
$Report = Join-Path (Get-Location) (
    '.local\standards-review\rehearsal-' +
    [guid]::NewGuid().ToString('N') + '.json'
)
if (Test-Path -LiteralPath $Report) {
    throw 'Refusing to overwrite an existing report.'
}
& $Python -B .\tools\rehearse_standards_release.py `
  --report $Report
if ($LASTEXITCODE -ne 0) { throw 'Synthetic rehearsal failed.' }
Get-Content -Raw -LiteralPath $Report
```

The harness independently refuses an existing report. Its expected contract is nine matching outcomes, including the four intentional negative calls, and a dry-run stub rather than model training. Random identifiers, timestamps, disposable signatures and temporary hashes may differ between repetitions. Compare the exact source selection and expected behavior rather than requiring every incidental value to equal the retained report. [E04, E08]

The retained suite had 47 passes and one link-capability skip. A new environment may execute the link assertion, but a later observation must be recorded as a new run rather than merged into the September total. Preserve each new failure or skip and its reason. Do not weaken a condition or copy an earlier PASS into a new report. Appendix C gives the capability boundary. [E03, E25, E27]

### 7.4 Historical reproduction and availability

Reproducing a historical model run or the full July frozen candidate is a different exercise. It requires the identified original source, applicable permissions, exact materials and appropriate environment. The compact source ledger is a locator and integrity aid, not a substitute for the absent datasets or a rights clearance. [E19, E22, E23]

The 6 September editorial comparison checked the eleven rehearsal source hashes and the 62 ledger references locally. It did not execute new tests or reopen the original model outputs, CSC-generation inputs, complete tagged freeze bundle or July export trees as full sets. Appendix A and the external crosswalk make that source-strength limit inspectable. The current manuscript pair and historical E01 reference remain distinct objects.

# 8 Limitations and review disposition

## 8.1 Technical limits

The evidence is bounded to the tested fixtures and recorded environment. Single-scope authorization does not implement one-time consumption, replay prevention, revocation infrastructure, key custody, trusted timestamps, professional identity verification or automatic assessment of free-text conditions. A valid signature is not proof that the identified reviewer had legal or organizational authority. [E07, E08]

Negative tests establish rejection for their specified cases. They do not prove absence of every filesystem side effect, complete parser safety, protection from every attack or exhaustive path portability. The historical symbolic-link skip and the later operator-reported pass have different evidence strength; no non-elevated pass or new full-suite outcome is established. The synthetic import/socket guard is not a general network-isolation guarantee for arbitrary future code. [E03, E04, E27]

No model-quality, inference-latency, throughput, cost, bias or generalization benchmark was performed for the September rehearsal or the 6 September documentary revision. No new v2 GPU fine-tune, complete three-path replay or CSC score is reported. Hosted CI status is not updated from the retained local evidence. [E04, E05, E09]

## 8.2 Legal-review and organizational boundaries

TR010 proposes qualified human review of identified legal questions where required by the adopting organization's policy, with separate organizational authorization of the named operation. This report records that project policy; it does not assert a universal legal duty to retain counsel for every AI experiment. AI assistance may organize evidence or drafting but is not accepted as the required practitioner approval under the proposed profile. [E01, E18, E19]

The worked example is explicitly fictional. Its actual status is `HOLD_PENDING_REAL_REVIEW`, and its organizational disposition is `NOT_AUTHORIZED_FOR_REAL_OPERATION`. There is no actual practitioner identity, verified practising credential, legal opinion or real authorization in that example. The current runtime does not consume the illustrative review record. It must not be relabelled as evidence that a lawyer reviewed this release. [E18]

The inherited licence and notices remain in place. External legal-review drafts remain separately identified rather than silently adopted. This report does not resolve ownership, standards-expression permissions, jurisdictional exceptions, confidentiality, reliance or distribution authority. Final manuscript status does not supply legal clearance for rights-dependent publication. [E02, E19]

## 8.3 Standards contribution

The retained evidence supports a bounded standards contribution: explicit checks on recorded permission and package integrity, a reproducible synthetic demonstration, and a comparison with earlier lifecycle results. The compact v2.0 Standards Edition reports that contribution; it does not claim lawyer review, independent certification or production readiness.

Further evaluation should examine the recorded technical gaps, independently reproduce the applicable tests and obtain original artifacts before making stronger historical claims. Rights-dependent use and publication remain separate decisions for the responsible people. The historical milestone sequence is not a new approval process for this study.

The two-paper structure separates the method from its verification. APOD-TR-010 provides the design and governance account; APOD-TR-010-R provides the observed results, reproduction route and limitations. The preserved ledgers, new companion-identity crosswalk and History and Provenance Notice retain the evidence identities without making manuscript drafting history part of the research claim.

**Conclusion.** Q-Pathformer Enterprise v2 provides a tested technical boundary for recorded authorization and package integrity. The evidence supports that bounded implementation claim, not automated governance authority. Historical results retain their original scope; new operations require their own current evidence and separately recorded authority.

## Appendix A Evidence catalogue

E01-E26 retain 26 groups and 60 file references. E27-E28 add two groups and two references. Their preserved ledger companions are `source_ledger_E01_E26.json` and `source_ledger_E27_E28.json`. The full paths, original hashes and current local comparisons are in `EVIDENCE_AVAILABILITY_CROSSWALK.json`. These records describe the inspected local evidence set. Local presence alone does not establish inclusion in a distributed archive.

On 6 September, all 62 recorded paths were present in the inspected local workspace and 61 matched the recorded bytes. E01 identifies an older R04 companion identity: its recorded SHA-256 begins `a6ceb060ad18`, while the later revision at that path begins `5c47b2d4591c`. These abbreviations are only reading aids; full hashes are in the crosswalk. Neither is the new compact main paper. The old ledger remains unchanged, and `MANUSCRIPT_REVISION_HANDOFF.json` identifies the earlier paper pair, not the current manuscripts.

In the table, "Exact" means present locally with the recorded bytes, not independent authentication of every assertion. "Different revision" means the E01 path contains a later manuscript. The table does not assert that every listed item is distributed with this edition. The concise locator identifies a source family; the historical machine crosswalk records the exact paths and hashes.

**Table A1 Source groups and local availability**

| Group | Purpose and evidence period | Local state | Concise locator |
|---|---|---|---|
| E01 | Companion paper APOD-TR-010; September documentary. | Different revision | TR010 earlier R04 manuscript |
| E02 | History and provenance notice; September documentary. | Exact | history_notice.md |
| E03 | September unit test log; September observed/source. | Exact | consolidation unit_tests.txt |
| E04 | September synthetic CLI rehearsal; September observed/source. | Exact | consolidation rehearsal.json |
| E05 | September recorded environment; September observed/source. | Exact | standards_local environment.json |
| E06 | September source captures; September observed/source. | Exact | Local inventory and SOURCE_CAPTURE |
| E07 | Current implementation and tests; September observed/source. | Exact | scripts schemas tools and unit tests |
| E08 | Current reviewer commands and pins; September observed/source. | Exact | Reference runbook and release pins |
| E09 | September consolidation checks; September observed/source. | Exact | Consolidation selection records |

| Group | Purpose and evidence period | Local state | Concise locator |
|---|---|---|---|
| E10 | July results and v1.0 comparison source; July historical. | Exact | TR010_DRAFTING v1.0 comparison |
| E11 | July deterministic export rehearsal; July historical. | Exact | M06 export rehearsal report |
| E12 | July corrected M10; July historical. | Exact | M10 remediation rerun report |
| E13 | July corrected M11; July historical. | Exact | M11 rebuild and verification |
| E14 | July M12 review and remediation scope; July historical. | Exact | M12 human and technical records |
| E15 | Historical v1.0 CSC qualification; July historical. | Exact | M08 qualification and manifest |
| E16 | Historical vendor equivalence and selected subset; July resolution and September selection. | Exact | STANDARDS_SUBSET and M09 resolution |
| E17 | Separate CSC governance snapshot contract; Historical design only. | Exact | Historical snapshot contract |
| E18 | Illustrative legal review; September documentary. | Exact | legal-review illustrative fixtures |
| E19 | Legal status and external references; September documentary. | Exact | Notices and external references |
| E20 | Local documentary provenance; September documentary. | Exact | R04 table-edit provenance |
| E21 | Windows dry-run investigation; September observed/source. | Exact | Windows dry-run investigation |
| E22 | Historical data selection boundary; September documentary. | Exact | evidence/historical_v1 README |
| E23 | Retained July publication records; July historical. | Exact | Historical inventory and checksums |
| E24 | Historical milestone sequence; July historical. | Exact | TR010_DRAFTING release process |
| E25 | Targeted Windows symbolic link recheck; September observed/source. | Exact | windows_symlink_recheck report |
| E26 | Standards review purpose and paper progression; September documentary. | Exact | Standards-review development plan |
| E27 | Operator report; September 6 record; execution time unrecorded. | Exact | operator_reported_symlink_pass |

| Group | Purpose and evidence period | Local state | Concise locator |
|---|---|---|---|
| E28 | Requirement sources; September documentary. | Exact | requirements_source_inventory |

Local presence of the retained July reports is not presence of their entire frozen candidate. Likewise, E15's qualification and manifest are not a newly generated set of CSC scores, and E18's legal fixture is not an actual legal engagement. E27 is operator-supplied, without a captured exact execution timestamp. E28 is requirement-source documentation, not a complete historical ML lock. [E10-E28]

## Appendix B Historical July release milestones

M means milestone: a checkpoint in preparing the release, not a model lifecycle state. This table condenses the retained July release-process narrative. Its arrows explain sequence, not permission. It does not restart the milestones or introduce a new gate. [E24]

**Table B1 Historical milestone sequence and limits**

| Sequence | Purpose in the July record | Boundary |
|---|---|---|
| M00 -> M01 | Agree scope and decision rules; plan the paper. | Unknown results are not fabricated. |
| M02 -> M03 | Inspect sources, rights questions and dependencies; select materials. | Selection is not legal clearance. |
| M04 -> M05 | Test copying rules; authorize the exact source baseline. | Source identity is not a public release. |
| M06 -> M07 | Compare two export targets; create the controlled candidate. | Equality applies to the recorded source and export rules. |
| M08 -> M09 | Qualify historical CSC evidence; prepare notices, docs, pins and CI definitions. | Sparse historical evidence and the bounded equivalence resolution remain qualified. |
| M10 -> M11 | Verify the candidate; write its ordered inventory and checksums. | Corrected M10/M11 PASS concerns that historical object. |
| M12 | Obtain the human release disposition. | Initial HOLD; corrected object ready for reconsideration, not a newly authorized M12 PASS. |
| M13A -> M13B -> M13C | Separately authorized local release, distribution and external publication/paper binding. | Not authorized or performed in the retained corrected-candidate record. |
| M14 | Separately versioned CSC Assurance Record work. | Future work, not an implemented scorer or automatic continuation. |

The bounded return loop was **M12 HOLD -> authorized M09 fixes -> M10 retest -> M11 rebuild -> M12 awaiting newly authorized review**. The corrected M11 success did not supply the next human decision. The later July 21 documentary context described in section 3.5 is not another completed pass through this loop.

# Appendix C System requirements for TR007 to TR010

Requirements depend on the activity, not merely the paper number. Reading these papers needs a compatible viewer, not a model runtime. The current synthetic route and the historical enterprise routes have different inputs and dependencies. The profiles below are source-derived conditions, not a certified platform specification. [E08, E28]

## C.1 Current synthetic verification

Use PowerShell 7 for the documented Windows commands and the selected existing Python 3.11 environment from section 7.1. Python 3.11.9 and the captured platform string `Windows-10-10.0.26200-SP0` identify the retained environment; they do not establish that every other platform was tested. Keep the eight dependency pins in section 2.1 unchanged. The selected source needs its scripts, schemas, tests, fixtures, backend profile and provenance records, with permission to create local temporary directories, child processes and a new report. [E05, E08]

The synthetic harness uses disposable authorization and sealing keys, JSON Schema validation and a dry run. GPU, CUDA, Ollama, TinyLlama, Docker and a GitHub account are not required for this route. Dependencies must already be prepared; the rehearsal itself does not need network access. Its import/socket guard is bounded, not a general operating-system sandbox. Ordinary checks do not require blanket administrator access. [E04, E07]

## C.2 Historical lifecycle and signal profiles

TR007 reproduction needs the identified lifecycle implementation, inputs, dataset passport, routing/lineage schemas and applicable authorization context. The inspected passport validation uses `jsonschema`. The source-first enterprise paths also need their actual source files, matching profiles and governance references. Path 1's PDF reader needs `pypdf`; its inspected DOCX reader uses ZIP/XML rather than installed Word. Path 3 needs the selected client JSON or freeze bundle and adapter profile. A missing dataset is not repaired by installing an LLM. [E28]

For TR008-related signal examples, the source-first Path 2 route consumes the identified freeze/bridge inputs without Ollama. The different runtime route requires a reachable local Ollama service, the selected `gpt-oss:20b` model, prompt/session identifiers and routing references; its default service endpoint is localhost port 11434. That prompt route does not automatically read `01_source`. Model storage and memory depend on the selected artifact and context; no universal minimum is established by the inspected records. [E28]

## C.3 Historical CSC and model profiles

Historical TR009 CSC generation needs the scorer/mapper, selected v1a catalogue and profiles, evidence schema and evidence JSON, plus any identified blueprint or scenario overlay. Use `jsonschema` for schema-validated scoring; the inspected historical scorer can otherwise omit that validation. No LLM, GPU or production certificate service is required for those local examples. Preserve coverage, observation windows, direct/proxy distinctions and exclusions. Missing telemetry is an evidence gap, not a value to invent. [E15, E17, E28]

Historical Path 1 or source-first Path 2 LoRA reproduction additionally needs its exact training package, backend profile, `TinyLlama/TinyLlama-1.1B-Chat-v1.0` weights/tokenizer and compatible `torch`, `transformers`, `datasets`, `peft` and `accelerate`. The inspected April summaries record CUDA. Reproducing that device profile needs a

compatible driver and CUDA-enabled PyTorch; auto-device selection does not prove CPU equivalence. The lightweight TR010 pins do not supply a complete historical ML lock or universal RAM/VRAM/disk minimum. No model download or historical training is needed to review the synthetic result. [E28]

### C.4 Symbolic link capability and separate observations

The exporter test creates a real link and checks that it is rejected as a source. A capability failure while creating the link occurs before that rejection assertion. Supported Windows Developer Mode or suitable privilege/elevation can permit link creation; the operator's use of both does not make both universal requirements. Apply organizational device policy before changing settings. No setting is changed by this manuscript. [W2, W3]

**Table C1 Separate link capability observations**

| Observation | Reported result | Limit |
|---|---|---|
| September 5 consolidation suite | 48 listed; 47 passed; one link-capability skip; zero failed. | Not a new all-pass suite. [E03] |
| Separate September 5 targeted recheck | Zero passed; one skipped; zero failed. Native probe reported error 1314. | Not a full-suite rerun; the earlier suite did not capture this probe error. [E25] |
| Later operator-supplied console output | One targeted test passed in 0.010 seconds; Developer Mode and elevated PowerShell reported. | Exact execution timestamp and unelevated pass not established; prior logs unchanged. [E27] |

For a separately authorized capability check, continue with the same workspace and selected `$Python` from section 7.1:

```
& $Python -B -m unittest discover `
  -s .\tests\unit `
  -p test_export_v2_public_release.py `
  -k test_source_symlink_is_rejected_when_supported `
  -v
if ($LASTEXITCODE -ne 0) { throw 'Targeted check failed.' }
```

Retain the complete output, including any skip. A zero exit with a skip is not proof that the assertion executed. A scan finding no links is also not that assertion. Remote Device Portal and Device Discovery are unnecessary for this test. Word is needed only for the manuscript pagination workflow, not Python execution. Git publication, signing, archive sealing and operational permission remain separate activities. [E25, E27, E28]

**[W1] Python 3.11 Windows launcher and virtual environments.** Official documentation.

**[W2] Python 3.11 os.symlink.** Official symbolic-link documentation.

**[W3] Microsoft developer settings.** Official settings guidance.

These references support environment interpretation, not the Q-Pathformer test outcomes. Actual local requirement-source identities remain in E28.

# References

## Published background

The foundational APOD-TR-002 and APOD-TR-006 Version E / E.2 entries are described in the companion TR010 bibliography. Reference identifiers B4-B10 are shared across the pair.

The entries below distinguish paper identifiers from release dates and package locators. A tagged file location is not an independent validation of that paper's datasets. Exact edition details not established by the inspected records are not invented.

**[B4] APOD-TR-007, Governed Dataset Onboarding and Lifecycle Routing.** Tagged paper, distributed with v1.0-qpath-dataset-lifecycle. The supplied reference audit reports the release date as 12 March 2026.

**[B5] APOD-TR-008, Interaction-Role Signal Exchange.** Paper at the enterprise v1.0 tag. Its historical scope and notices remain distinct from the new paper pair.

**[B6] APOD-TR-009, Access PoD CSC Session Model and Execution.** Tagged paper.

**[B7] APOD-TR-009-R, Access PoD CSC Results.** Tagged results paper. B5-B7 are distributed with the enterprise v1.0 release, reported by the supplied reference audit as published on 18 April 2026 and anchored to commit `0fa94ecf47d698a894bc868545f110570d0dec50`.

**[B8] QPATH-ARCH-001, Q-Pathformer Multi-state Machine Learning and LLM Training.** Paper at the lifecycle tag. The tagged directory confirms the file title and location; no independent internal edition date or architecture benchmark is inferred.

**[B9] Access-PoD Artifact Pack v1a.** Published public-review pack, published on 28 December 2025. The publisher distinguishes its curated pack from automatically generated source archives and disclaims certification or regulatory approval. This is background for the historical CSC records, not a new certification scheme.

**[B10] Access-PoD governed dataset lifecycle release.** v1.0-apod-dataset-lifecycle. Its governance artifacts remain distinct from the Q-Pathformer execution tooling.

## Current v2 standards paper pair

**[P1] Anthony Siljeg. APOD-TR-010.** Fail-Closed Governed Dataset Lifecycles and Explicit Training Authorization for AI Training Systems. Compact standards edition, 6 September 2026. Final manuscript for the v2.0 Standards Edition; method and reusable transition protocol.

**[P2] Anthony Siljeg. APOD-TR-010-R.** Verification Results and v1.0 Comparison. Compact standards edition, 6 September 2026. Final manuscript for the v2.0 Standards Edition; bounded September results, historical comparisons and source catalogue.

The preceding DOCX and reading-copy PDF identities are recorded in `TR010_R02_EDITORIAL_HANDOFF.json`. Earlier handoffs and E01/E20 and edition ledgers retain their dated scope. Final publication PDF identities are recorded separately after conversion and review; neither paper embeds its own final hash in its body. `EVIDENCE_AVAILABILITY_CROSSWALK.json` is the dated crosswalk locator; the two historical ledger companions identify E01-E26 and E27-E28. Their inclusion with this paper pair is not established by these references.

### V2.0 Standards Edition release

Repository: `qpathformer/qpathformer-enterprise-v2.0-standards-release`.

Release tag: `v2.0-qpathformer-enterprise-standards`.

V2.0 Standards Edition release.

The repository and release tag identify this v2.0 Standards Edition, separately from the historical July candidate. The compact edition reports bounded synthetic verification, not full enterprise reproduction. These identifiers do not themselves confer rights clearance, independent assurance or operational authorization.

### Planned v3 enterprise destination not published

Repository: `qpathformer/qpathformer-enterprise-v3.0-release`.

Proposed tag: `v3.0-qpathformer-enterprise`.

Planned v3 enterprise destination not published.

The proposed enterprise scope combines the v1.0 foundation with v2 controls for defined end-to-end reproduction. It is not a verified v2 result. The Q-Pathformer organization directory and Access-PoD release catalogue provide navigation; they do not establish publication of the planned v3 release.

The 6 September 2026 reference audit records the historical tagged locations and the scope of the checks performed. No release ZIP, historical model output or full dataset was downloaded or re-executed for those reference checks. A release description, tag, commit, file hash and sealed archive digest are different identities. No new licence grant or legal clearance is inferred from a reference link.